Amazon Web Services

(AWS)

A Comprehensive Guide to Cloud Computing

Cloud Infrastructure | Core Services | IoT | Architecture | Pricing

Introduction to AWS

☁️ What is AWS?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, launched in 2006 by Amazon.com.

🌍 **Global Reach**	31 launched regions, 99 availability zones
🏢 **Users**	Over 1 million active customers worldwide
🚀 **Services**	200+ fully-featured services
📅 **Founded**	2006 – EC2 & S3 launched

🏆 Why AWS Matters

📈 **Market Leader**

33% cloud market share (2024)

⚡ **Innovation Speed**

Launch infra in minutes, not months

💰 **Cost Savings**

Pay only for what you use

🔒 **Enterprise Security**

99.99% SLA, 300+ compliance certifications

🌐 **Global Scale**

Deploy in any region instantly

🔧 **Full Ecosystem**

ML, IoT, Analytics, DevOps, and more

🌟 Used by Netflix, NASA, Airbnb, Samsung, GE, and 1+ million organizations globally

Characteristics of AWS

Scalability

Scale resources up/down instantly based on demand. Auto Scaling groups adjust EC2 instances automatically.

Elasticity

Dynamically provision and de-provision resources in real-time to match workload changes.

Reliability

Multiple availability zones redundancy ensures 99.99% uptime Service Level Agreement. Automatic failover keeps apps running.

Security

300+ compliance certifications. Encryption at rest and in transit. Identity & Access Management (IAM) for access control.

Cost-Effective

Pay-as-you-go model. No upfront hardware. Reserved instances save up to 75%.

Global Reach

31 geographic regions, 99 AZs. Deploy your app close to users anywhere in the world.

High Performance

Custom silicon (Graviton CPUs), low-latency networking, and SSD-backed storage.

Managed Services

AWS handles patching, backups, and hardware. You focus on code, not infrastructure.

AWS Service Models: IaaS · PaaS · SaaS

IaaS

Infrastructure as a Service

Like renting land & building from scratch

You manage: OS, Runtime, Apps

AWS: Hardware, Network, Virtualization

AWS Services:

- EC2 – Virtual Machines
- VPC – Private Networking
- EBS – Block Storage
- S3 – Object Storage
- Route 53 – DNS

PaaS

Platform as a Service

Like renting a furnished apartment

You manage: Apps & Data

AWS: OS, Runtime, Middleware, Infra

AWS Services:

- Elastic Beanstalk – App Platform
- RDS – Managed Database
- Lambda – Serverless Functions
- ECS – Container Service
- CodePipeline – CI/CD

SaaS

Software as a Service

Like staying in a hotel – fully managed

You manage: Only your data

AWS: Everything else

AWS Services:

- WorkMail – Email Service
- Chime – Video Conferencing
- QuickSight – BI Dashboard
- Rekognition – Image AI
- Alexa for Business

Virtualization in AWS

🔧 AWS Nitro Hypervisor Stack

Guest VMs / Containers

EC2 Instances (t3, m5, c6g…)

Nitro Hypervisor (KVM-based)

Ultra-thin, hardware-accelerated

Nitro Cards (Custom Hardware)

Network, Storage, Security offload

AWS Nitro Security Chip

Cryptographic root of trust

Physical Host (AWS Data Center)

Custom ARM/x86 servers

📦 Virtualization Types

🖥️ Full Virtualization (HVM)

Hardware Virtual Machine mode — full OS isolation. Used by modern EC2 (m5, c5, r5).

🔌 Para-Virtualization (PV)

Guest OS aware of hypervisor. Older AMIs, faster I/O but limited to older instance types.

📦 Container (Docker/ECS)

Lightweight, shared kernel. AWS ECS, EKS, Fargate. No full VM overhead.

⚡ Serverless (Lambda)

No VMs visible to user. AWS manages microVMs via Firecracker VMM internally.

⚙️ Bare Metal (i3en)

Direct hardware access, no hypervisor. For HPC, SAP HANA, DB licensing.

Core AWS Services & Their Roles

EC2
Compute

Virtual servers in the cloud. Launch in seconds, pay per hour. Powers most web apps.

S3
Storage

Unlimited object storage. Store files, backups, videos. 99.999999999% durability.

RDS
Database

Managed SQL databases: MySQL, PostgreSQL, Oracle, SQL Server. Auto backup & scaling.

Lambda
Serverless

Run code without servers. Pay only per invocation. Auto-scales to millions of requests.

VPC
Networking

Isolated virtual network in AWS. Control subnets, routing, firewalls, gateways.

IAM
Security

Manage who can access what in AWS. Users, roles, policies. Zero trust foundation.

CloudWatch
Monitoring

Logs, metrics, alarms. Monitor EC2 CPU, set alerts, auto-remediate issues.

EKS
Containers

Managed Kubernetes clusters. Deploy containerized microservices at scale.

SNS/SQS
Messaging

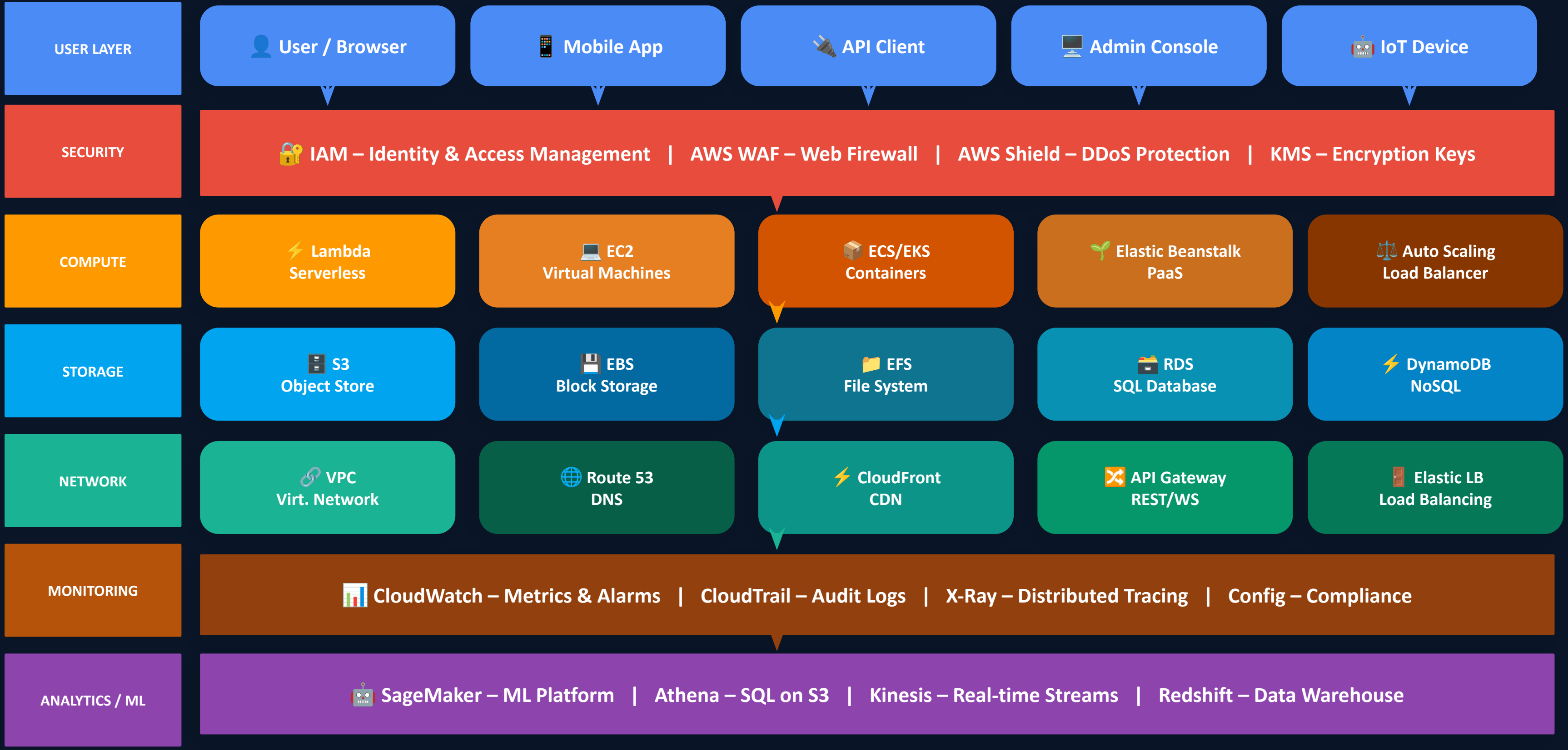
SNS: push notifications/pub-sub. SQS: message queues for decoupling services.

Route 53
DNS

Scalable DNS with health checks, failover routing, and latency-based routing.

AWS Functional Architecture

SECTION 6 – KEY TOPIC



Flow: User Request → IAM Auth → Compute Processing → Storage R/W → Network Routing → Monitored → Analyzed

Architecture Flow – Step by Step Walkthrough

1 User Request

User opens a browser or mobile app (e.g., Netflix on AWS). Request travels over the internet.

2 DNS Resolution

Route 53 resolves the domain name to the nearest CloudFront edge location for low latency.

3 CDN Edge Cache

CloudFront serves cached static content (images, CSS, JS) from 450+ edge locations globally.

4 IAM Authentication

AWS IAM validates API tokens/credentials. WAF inspects for SQL injection, XSS attacks.

5 Load Balancer

Elastic Load Balancer distributes traffic across multiple EC2/container instances in different AZs.

6 Compute Layer

EC2, Lambda, or ECS processes the request — runs business logic, queries database, calls APIs.

7 Storage & Database

Data read/written to RDS (SQL), DynamoDB (NoSQL), S3 (files), ElastiCache (in-memory cache).

8 Response + Monitoring

Response returned to user. CloudWatch logs metrics, triggers alarms. CloudTrail records all API calls.

🎬 Real Example: Netflix on AWS — User clicks 'Play' → Route 53 → CloudFront CDN → IAM → Load Balancer → EC2/Lambda → DynamoDB → CloudWatch

AWS IoT Services

SECTION 7 – KEY TOPIC

AWS IoT Core

Central cloud hub. Connects billions of devices via MQTT/HTTPS. Routes messages to AWS services.

AWS IoT Greengrass

Edge computing on device. Run Lambda locally, sync with cloud even offline. Smart factory use.

IoT Device Shadow

Virtual representation of device state. Query last known state even when device is offline.

IoT Analytics

Process & analyze IoT data. Filter noise, run ML models, generate dashboards via QuickSight.

IoT Events

Detect patterns and trigger actions. E.g., alert if temperature > 80°C for 5 consecutive readings.

IoT SiteWise

Industrial IoT monitoring. Collect data from factory OPC-UA equipment, model asset hierarchies.

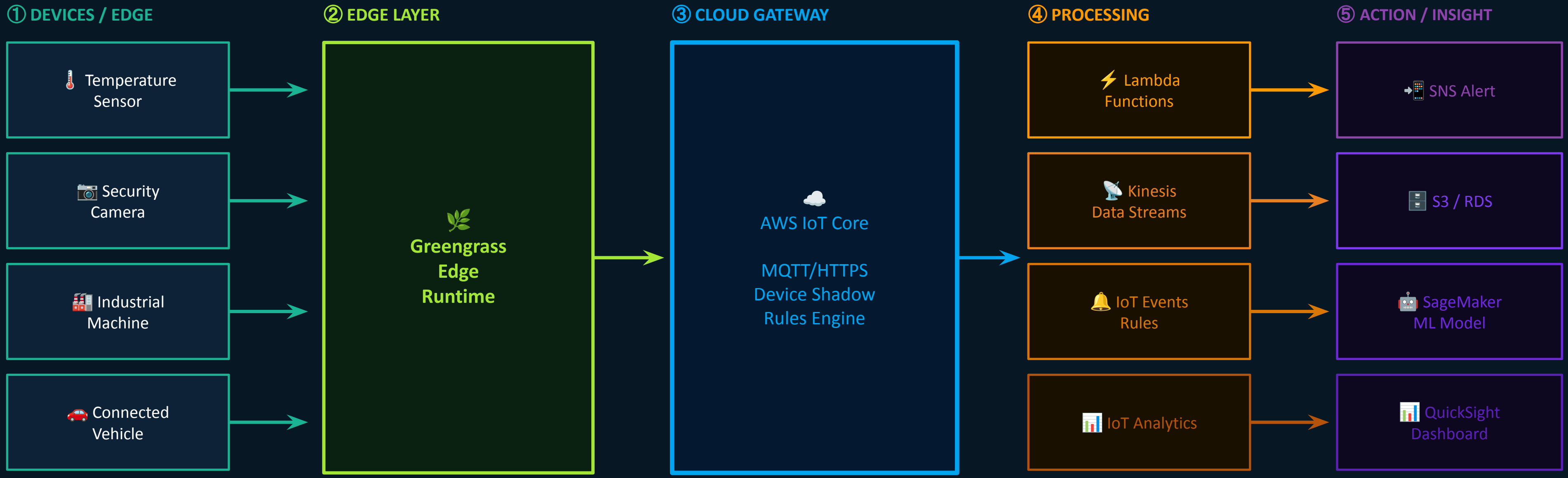
IoT TwinMaker

Create digital twins of physical systems. 3D visualization of buildings, factories, equipment.

FreeRTOS

Real-time OS for microcontrollers (MCUs). Securely connect tiny devices to AWS IoT Core.

AWS IoT Architecture Flow



Protocol: MQTT (lightweight, publish/subscribe) → Secure TLS encryption → Bidirectional Device–Cloud communication

🔐 All IoT traffic encrypted with TLS 1.2+ | X.509 certificate-based device authentication | IAM policies control data access

Real-World IoT Use Cases on AWS:

🏠 Smart Homes: Alexa/Ring → IoT Core → Lambda | 🌾 Agriculture: Soil sensors → Greengrass → Analytics → Irrigation trigger | 🏥 Healthcare: Vital monitors → IoT Core → SageMaker anomaly detection → SNS alert to doctor

AWS IoT Real-World Use Cases

🏠 Smart Home

📡 Devices:

Alexa Echo, Ring Doorbell, Smart Thermostat

🔄 Data Flow:

Device → IoT Core → Lambda → DynamoDB

✅ AWS Features Used:

- Voice control via Alexa Skills Kit
- Ring video stored on S3, analyzed by Rekognition
- Nest-like thermostat: ML predicts preferences
- Automation rules via IoT Events

🏢 Amazon Ring, Philips Hue, Google Nest on AWS

🌾 Smart Agriculture

📡 Devices:

Soil moisture, temperature, humidity sensors

🔄 Data Flow:

Sensors → Greengrass → IoT Core → Analytics → Irrigation

✅ AWS Features Used:

- Greengrass runs ML model at edge field
- Kinesis streams real-time crop data
- SageMaker predicts yield & disease risk
- Auto-trigger irrigation via Lambda + IoT Events

🏢 John Deere, Farmers Edge use AWS IoT

🏥 Healthcare IoT

📡 Devices:

ECG monitors, glucose sensors, wearables

🔄 Data Flow:

Device → HIPAA-compliant IoT Core → HealthLake → Alert

✅ AWS Features Used:

- HIPAA-compliant data pipeline
- Anomaly detection: SageMaker flags arrhythmias
- SNS sends real-time alert to nursing staff
- All data encrypted with KMS keys

🏢 GE Healthcare, Philips eICU on AWS

AWS Pricing Model & Cost Structure

⚡ On-Demand

Baseline Cost

Pay per hour/second, no commitment

Maximum flexibility. Use any time. Stop and you stop paying. Perfect for dev/test, unpredictable workloads.

📌 *Example: EC2 t3.medium: ~$0.0416/hour*

📋 Reserved Instances

Up to 72% off

1–3 year commitment

Up to 72% savings vs On-Demand. Ideal for predictable, steady-state workloads like production databases.

📌 *Example: 1-yr RDS db.t3.medium: ~$0.067/hr*

🎯 Spot Instances

Up to 90% off

Bid on spare EC2 capacity

Up to 90% savings. AWS can terminate with 2-min notice. Best for batch jobs, ML training, video rendering.

📌 *Example: c5.xlarge spot: ~$0.04 vs $0.17*

💰 Savings Plans

Up to 66% off

Commit to $/hr spend for 1-3 years

Flexible than Reserved — applies across instance families & regions. Compute and EC2 Savings Plans available.

📌 *Example: Compute Savings Plan: auto-applied*

🎁 AWS Free Tier: 750 hrs/mo EC2 t2.micro + 5 GB S3 + 25 GB DynamoDB + 1M Lambda requests — FREE for 12 months (new accounts)

Resource Provisioning Methods

🖱️ Manual Provisioning

AWS Management Console

Web UI — point and click to launch EC2, configure S3, etc. Beginner-friendly.

AWS CLI

Command-line tool. Scriptable, faster than console for repetitive tasks.

AWS SDKs

Programmatic API via Python (boto3), Java, Node.js. Integrate AWS into your app.

Limitations

Manual = error-prone, not reproducible, slow for large environments.

⚙️ Infrastructure as Code (IaC)

AWS CloudFormation

Define infra in YAML/JSON templates. AWS provisions it automatically. Version-controlled.

AWS CDK (Cloud Dev Kit)

Write infra in Python/TypeScript/Java. CDK compiles to CloudFormation.

Terraform (HashiCorp)

Open-source IaC. Multi-cloud (AWS+Azure+GCP). HCL language. Widely used.

Benefits

✅ Repeatable ✅ Version controlled ✅ Fast deployments ✅ Disaster recovery

User Interfaces & Programming Support

🖥️ AWS Management Console

Web-based GUI accessible at console.aws.amazon.com. Visual dashboard for all AWS services. Best for beginners and occasional management tasks.

- `Point-and-click interface`
- `Service health dashboard`
- `Cost Explorer visualization`
- `Role-switching across accounts`

💻 AWS CLI (Command Line)

Powerful command-line tool for all AWS operations. Install once, control everything. Supports scripting and automation in bash/PowerShell.

- `aws ec2 describe-instances`
- `aws s3 cp file.txt s3://bucket`
- `aws lambda invoke function`
- `Supports --profile for multi-account`

⚙️ AWS SDKs (Programmatic)

Native language libraries to call AWS APIs from your application code. Auto-handles auth, retries, and pagination.

Supported Languages:

Python `boto3`	**JavaScript** `aws-sdk / @aws-sdk`
Java `aws-java-sdk-v2`	**Go** `aws-sdk-go-v2`
Ruby `aws-sdk-ruby`	**.NET/C#** `AWSSDK.Core`

💡 **Pro Tip: Use Console to explore, CLI to automate, SDKs to integrate AWS into your application code**

AWS at a Glance

Everything we covered

1 **Introduction**
AWS = #1 cloud, 200+ services, 31 regions

2 **Characteristics**
Scalable, elastic, reliable, secure, cost-effective

3 **Service Models**
IaaS (EC2/S3), PaaS (Beanstalk/RDS), SaaS (Chime)

4 **Virtualization**
Nitro Hypervisor, HVM, Containers, Serverless

5 **Core Services**
EC2, S3, RDS, Lambda, VPC, IAM, CloudWatch

6 **Architecture**
User⟶IAM⟶Compute⟶Storage⟶Network⟶Monitor

7 **IoT on AWS**
IoT Core, Greengrass, Analytics, Real-world cases

8 **Pricing**
On-Demand, Reserved (72% off), Spot (90% off)

9 **Provisioning**
Console, CLI, CloudFormation, Terraform, CDK

10 **Interfaces**
Console, CLI, SDKs (Python boto3, Java, Node.js)

Thank You! | AWS — Build anything. Scale globally. Move faster.